

CSCI 6420 Final Project (100 Points)

Dear Students,

Firstly, by attempting this project you agree to the condition that all the risks associated with the incorrect implementation is your responsibility.

For this project, please follow the steps below to complete it successfully. Please remember that you will have to take the screen shots in each and every step and document every step as a part of the report. Your project screenshots have to be very clear so that I can see each step very clearly. If the image is blurred, you will have to use special software packages like “GIMP” to make it look nicer. Each task has points mentioned next to it. Your report will be in the same order as the parts and has to be submitted in the following format: **Firstname_Lastname_FinalProject.pdf**.

Task 1) Using Datasets from Zybooks. (30 Points)

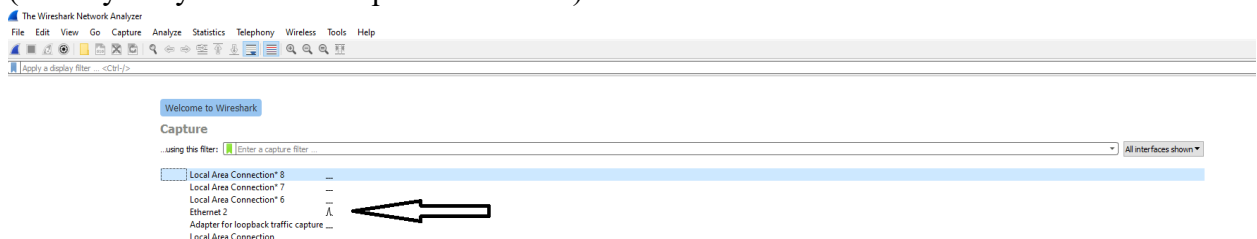
In order to complete this task successfully, please do the following:

- In the section 13.1, you have datasets in the CSV format. Choose 5 datasets from that list and Convert them into ARFF (WEKA readable) format.
- After the conversion, repeat steps 1-4 from HW5, Task 1.
- Also, complete step 5 from HW5 for each dataset listed here.

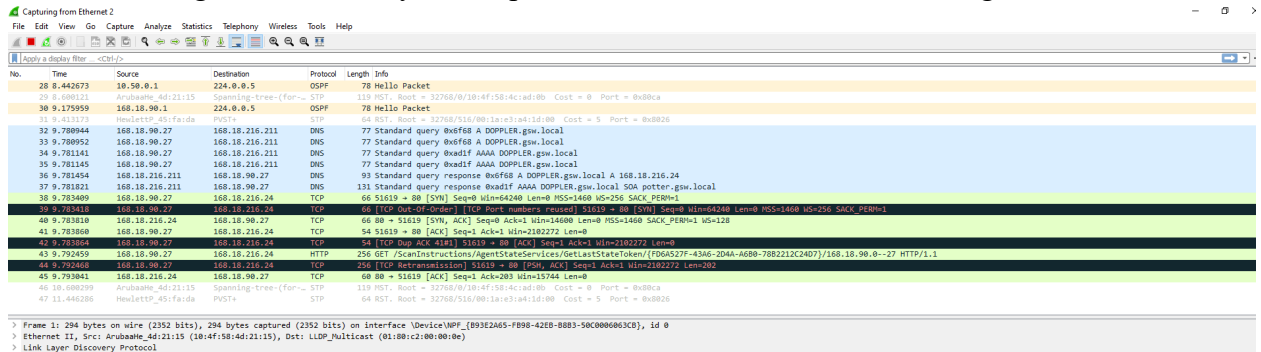
Task 2) Creating your own Dataset. (30 Points)

Till now you have been using datasets that been already created for you. For this task, you will create your own dataset. There are many ways to create a dataset. This is just one of them. Please follow the steps below to create the dataset successfully:

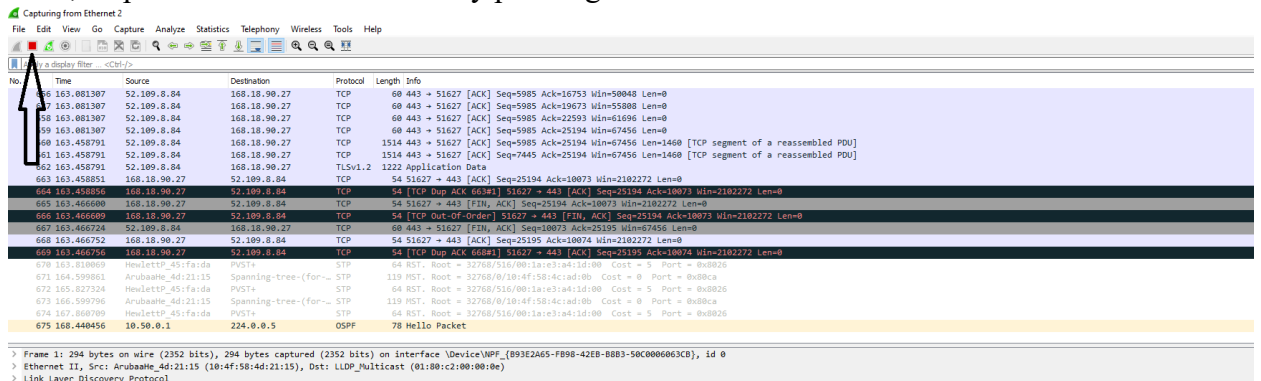
- Download and Install “Wireshark”.
- Launch Wireshark on a Network interface that captures the traffic as shown below.
(Usually it's your WIFI Adapter or Ethernet)



- After launching Wireshark on your adapter the screen should look something like this:

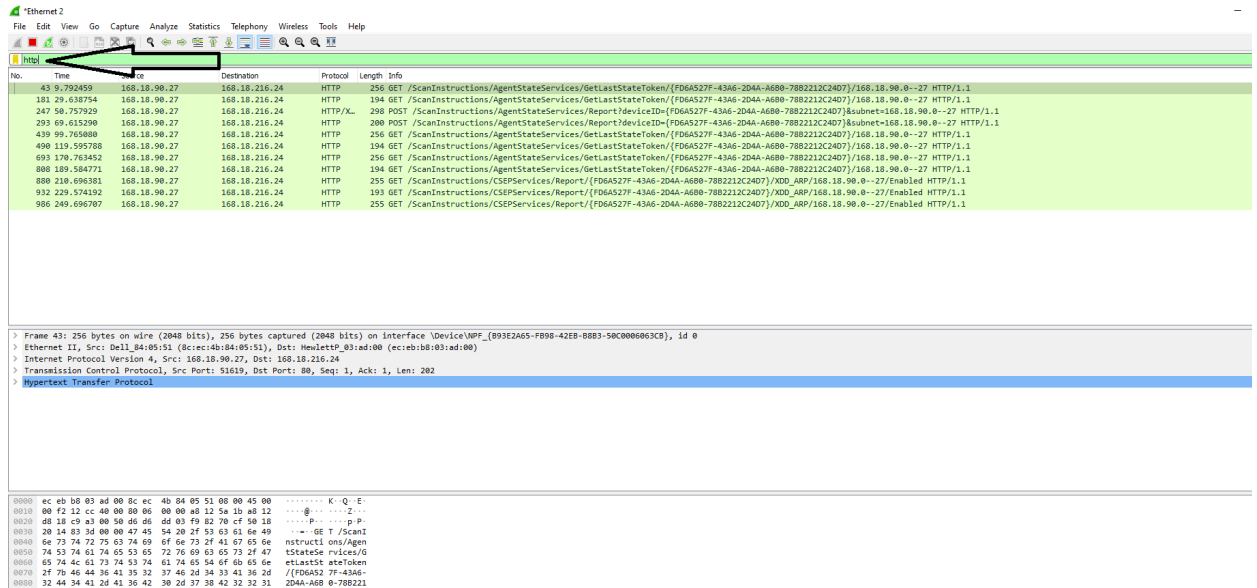


- Now, open Firefox or chrome and launch multiple tabs with different websites. E.g. Open GaView on one tab, watch your favorite YouTube video on another tab etc. Do this for five minutes.
- Now, stop the traffic in Wireshark by pressing the red icon.



- Filter just the “http” traffic as shown below.





- After filtering, convert the file into CSV and then into ARFF. (This is a very interesting step. I will leave it to you to get the output successfully.)
- Save the ARFF file as “Sample.arff” on your Desktop.
- Load the file into WEKA and perform the steps 1-5 from HW5, ask 1.
- Comment on the difference in the output from the previous tasks and the reason why you think the output differs.
- **Also, turn in your dataset on GaView (separately).**

Task 3) Deeplearning using WEKA. (40)

Since this class is about data and algorithms, it would be unfair to graduate from this class without learning about “Deeplearning”. Please complete the following tutorial (use the second link provide below) to complete Deep learning using WEKA. Just FYI, I have tested some of this on my end and it works pretty well. (You don’t need coding for this. Please follow the steps provided in the tutorial and use the GUI version only).

<https://deeplearning.cms.waikato.ac.nz/>

<https://deeplearning.cms.waikato.ac.nz/examples/inference/>